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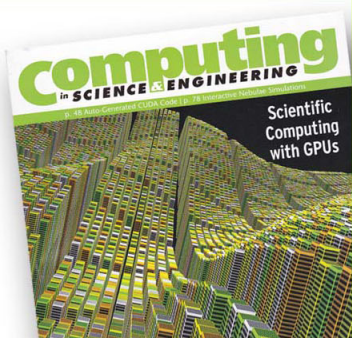
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Energy transfer in NH_3 -He collisions^{a)}

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The electron gas intermolecular potential for NH_3 -He developed previously is shown to be inadequate for a quantitative description of energy transfer in this system by comparing with available experimental data—pressure broadening, double resonance, and beam scattering. A new potential is constructed by combining the Hartree-Fock results of Davis, Boggs, and Mehrotra with semiempirical long-range induction and dispersion terms. The resulting potential is shown to be in harmony with beam scattering and pressure broadening measurements. It is found that the quadrupole-induced-dipole interaction, which has been invoked previously to account for pressure broadening in this system, is not, in fact, as effective as the short-range anisotropy. Although results are in general accord with double resonance data, a few unresolved discrepancies remain. It is suggested that these are due to improper consideration of the dependence of cross sections on degenerate magnetic quantum levels in the experimental analysis. Using this new intermolecular potential state-to-state rate constants are presented for temperatures to 300 K.

I. INTRODUCTION

Analysis of radiofrequency observations of interstellar ammonia requires rate constants for collisional excitation by H_2 and He. It is not currently possible to obtain the necessary state-to-state rate constants experimentally, especially at the low temperatures relevant to astrophysics. Some related information is available, however, from various experiments: pressure broadening, double resonance, and molecular beam differential scattering. In principle, the collision rates can be computed theoretically. For excitation by He atoms, such calculations are feasible with present capabilities; for excitation by H_2 , a proper treatment is probably prohibitive. However, it has been argued from both experimental evidence and theoretical considerations that excitation by H_2 —especially by cold para- H_2 —is similar to excitation by He.

The present work considers the available experimental and theoretical studies of NH_3 -He collisions and presents results from new calculations. The available experimental data are used to assess the accuracy of theoretical results.

II. EXPERIMENTAL DATA

A. Pressure broadening

A number of measurements have been reported for collision induced pressure broadening of the microwave inversion levels of ammonia, both self-broadening and foreign gas broadening by a variety of gases including He and H_2 .¹ It is difficult to achieve accuracy of better than 5%–10% in such experiments. However, within this constraint various experimental values are consistent. Pressure broadening coefficients can be expressed in terms of an equivalent cross section. The pressure broadening cross section for NH_3 -He is found to be about $18 \text{ \AA}^2$ at 300 K with little dependence on rotational level.

For NH_3 - H_2 the cross section is found to be about $30 \text{ \AA}^2$ at 300 K, again with little dependence on rotational level. Recently, Oka² has made measurements of broad-

ening of the $(j, k) = (3, 3)$ and $(2, 2)$ levels of NH_3 by para- H_2 as well as normal- H_2 , and at 207 K and room temperature. Assuming that broadening by H_2 with $j \neq 0$ is independent of j , he infers the following pressure broadening cross sections for the $(2, 2)$ line:

Temperature, K	H_2 $j=0$	H_2 $j \neq 0$
294	$21 \text{ \AA}^2$	$31 \text{ \AA}^2$
207	$22 \text{ \AA}^2$	$37 \text{ \AA}^2$

Values for the $(3, 3)$ line differed by less than 5%. This supports the assumption that cold para- H_2 ($j=0$) is very similar to He, whereas H_2 with $j \neq 0$ gives larger cross sections due to the long-range dipole-quadrupole interaction.

The pressure broadening cross section is given approximately as the average over the two spectroscopic levels of the total inelastic cross sections. The calculations discussed below indicate that this is an excellent approximation for thermal energy NH_3 -He collisions. In analyzing NH_3 pressure broadening it is sometimes further assumed that the intradoublet cross section is much larger than cross sections to other doublets; i.e., the pressure broadening cross section is assumed to be essentially the collision cross section between the two spectroscopic levels, cross sections to other (j, k) levels being negligible by comparison. The reasoning behind this approximation is that the intradoublet energy spacings are quite small, about 0.8 cm^{-1} whereas the energy change to reach a different (j, k) pair is at least 10 times larger. While this may be a reasonable approximation when the scattering is dominated by long-range forces and glancing collisions (as is the case in self-broadening or broadening by other polar molecules) it is likely to be a poor approximation for the short-range collisions which dominate in NH_3 -He. In fact, the detailed calculations discussed below indicate that the intradoublet cross section accounts for less than about half of the NH_3 -He pressure broadening cross section at thermal energies.

B. Microwave double resonance

In double resonance experiments, one perturbs the equilibrium distribution in two molecular levels by applying a strong radiation field (the pump) at the resonance frequency. This perturbation is propagated to other molecular levels via collisions which is monitored by observing the change in intensity of a weak field (the signal) at the resonance frequency between two other levels. Oka³ has given a good review of the experimental techniques, the theoretical analysis, and results for several systems including extensive studies of NH₃.

In NH₃ it is convenient to choose one (j, k) inversion doublet for the pump levels and a second (j', k') doublet for the signal levels. To a first approximation, the relative change in intensity, $\Delta I/I$, in the signal levels, on applying power to the pump levels is given by (see Ref. 4)

$$\Delta I/I \approx (k_{ps}^{lu} - k_{ps}^{ll})/k_{tot}, \quad (1)$$

where k_{ps}^{lu} is the collision rate from the lower level of the pump doublet to the upper level of the signal doublet (this is equal to the rate from the upper level of the pump doublet to the lower level of the signal doublet), k_{ps}^{ll} is the rate from the lower level of the pump doublet to the lower level of the signal doublet (which is equal to the rate from the upper level of the pump doublet to the upper level of the signal doublet), and k_{tot} is the total inelastic rate out of each signal level. The total rate k_{tot} is similar to the rate measured by pressure broadening. A more careful derivation of Eq. (1), as given, e.g., in Ref. 3, shows that the rate in the denominator k_{tot} should count the rate between signal levels k_{ss}^{lu} twice. This would reduce the calculated magnitude of $\Delta I/I$ by some 20%–50%, but it would not change the *sign* of the effect, since this depends only on the relative size of lower-to-lower and lower-to-upper collision rates. In the present work Eq. (1) will be used in order to compare more directly with the earlier theoretical values.

Several assumptions are made in deriving Eq. (1), two of which deserve further attention.^{3,4} First, multiple collisions have been ignored. Thus the possibility that a population perturbation is carried by one collision from the pump (j, k) doublet to an intermediate (j'', k'') doublet and by a second collision to the signal (j', k') doublet has been ignored. It has been argued by Oka³ that the small magnitude of the first order (i.e., single collision) effect is adequate to justify ignoring higher order terms. Of course, given a complete set of theoretical rate constants it is possible to solve for the population transfer exactly, without this approximation; and Davis and Boggs⁴ have demonstrated from such calculations that the single collision assumption is accurate. A second point which needs to be considered is the treatment of the $2j+1$ degenerate magnetic sublevels (m quantum number). Astrophysical applications require only a degeneracy averaged cross section. The derivation of Eq. (1) has implicitly assumed an average cross section as well. However, as noted by Oka³ the pump radiation does not affect the different m levels equally. In particular, the $m=0$ levels are not pumped at all. Oka has argued that this will not affect results signifi-

cantly, especially for high j values, and under the assumption that collisions generally leave m unchanged or change it by only one unit. This last assumption, while reasonable for long-range glancing collisions, is probably not justified for the short-range collisions in NH₃-He and NH₃-H₂. It is possible to compute the m dependence of cross sections in a scattering calculation. However, this involves a significant increase in the computational effort; further, approximate scattering formalisms, such as the coupled states method, may not predict the m dependence correctly. To date no calculations have been performed for the m dependence in NH₃-He collisions.

C. Molecular beam scattering

Differential cross sections have been measured for NH₃-He and NH₃-H₂ collisions by Bickes *et al.*⁵ Such measurements are thought to sample the isotropic part (i.e., the average over orientations) of the potential energy surface.⁶ The data were fit to a Lennard-Jones (12,6) potential which has two adjustable parameters, the well depth ϵ and the position of the minimum r_m . Results can be summarized as

	σ , bohr	r_m , bohr	$(4\pi)^{1/2} \epsilon$, cm ⁻¹
NH ₃ -He	5.84	6.55	64
NH ₃ -H ₂	5.84	6.55	277

The assumed form for the potential forces the position of zero energy σ to equal $2^{-1/6} r_m$. It should be noted that earlier results for NH₃-H₂ from similar experiments by Kuppermann *et al.*⁷ implied a value for r_m some 10% larger than the value given above.

III. PREVIOUS THEORETICAL STUDIES

The theoretical determination of collision rates can be separated into two problems, (1) calculation of the intermolecular forces or potential energy surface, and (2) calculation of scattering dynamics on this surface.

Green⁸ has considered the excitation of NH₃ by collisions with He. The potential was obtained from a theoretical electron gas (EG) model developed by Gordon and Kim.⁹ It can be noted that the positions of the zero and minimum of energy (σ and r_m) in the spherical part of the EG potential are some 5% smaller than values inferred from beam scattering experiments, while the well depth (ϵ) is nearly three times larger. Using this potential, accurate, converged close coupling (CC) scattering calculations were presented at a few collision energies. It was also shown that the computationally simpler coupled states (CS) scattering approximation provided good accuracy compared with the CC results. Ammonia was treated as a rigid rotor in this study; it was argued that the inversion motion, which is very slow on the time scale of rotational motion, is not important.

Subsequently, Davis and Boggs⁴ employed the same EG potential to perform semiclassical scattering calculations which were shown to agree closely with the quantum CC values. These authors also demonstrated rigorously that the inversion motion of NH₃ can be ig-

nored for this process. Davis and Boggs calculated cross sections for para- NH_3 ($k=3n\pm 1$) at a number of collision energies and performed a Boltzmann average to get rates at room temperature. Using these rates the authors modeled the microwave double resonance experiments of Oka.³ It was found that theory was in rather good accord with experimental values for some cases, in particular those in which the pumped levels and the monitored (signal) levels did not have the same value for the k quantum number ($\Delta k=3$); however, for cases where the k quantum number was the same ($\Delta k=0$) agreement between theory and experiment was poor. Davis and Boggs suggested two likely sources for the disagreement: (1) inaccuracies in the potential, and/or (2) improper treatment of the degenerate magnetic sublevels.

In order to assess the accuracy of the EG potential for $\text{NH}_3\text{-He}$, Davis, Boggs, and Mehrotra¹⁰ have computed a Hartree-Fock (HF) potential. They found that the EG model provided a reasonable qualitative description for many aspects of the potential surface, but that significant quantitative discrepancies were evident. Scattering calculations using this surface were not performed.

IV. NEW THEORETICAL STUDIES

A. Pressure broadening with the EG potential

To provide a further test of the EG potential, we have used it to compute the pressure broadening cross section for the (1,1) doublet. The calculation employed the coupled states approximation (see, e.g., Ref. 8) and a relatively small basis set expansion consisting of the six levels with $j=1$ and $j=2$ (this will be designated CS/B6). The collision energy was fixed at 200 cm^{-1} ($\sim 300\text{ K}$). This gave a cross section of only $4.8\text{ \AA}^2$, more than three times smaller than the experimental value. While the relatively small calculation employed here is not converged, it is accurate enough (probably better than 20%) to indicate the inadequacies of the EG potential for predicting collision cross sections. This calculation used an essentially exact formulation to compute the pressure broadening cross section from the collisional S -matrices, and it is interesting to note that this was only about 3% larger than the approximate value obtained from the total inelastic cross section.

B. Pressure broadening with the HF potential

Since the EG potential appeared to be inadequate based on comparison with double resonance and pressure broadening experiments, it became of some interest to consider the accuracy of the Hartree-Fock potential computed by Davis *et al.*¹⁰ These authors provided the spherical harmonic expansion coefficients $v_{\lambda\mu}(R)$ at a small number of radial distances, $R=3(1)8$ bohr (cf. their Table V). To obtain continuous functions for use in scattering calculations these were interpolated in each interval as

$$v_{\lambda\mu}(R) = \pm \exp(aR + b), \quad (2)$$

where the sign was chosen to provide the correct sign for $v_{\lambda\mu}$ and a and b were chosen in each interval to fit the two end points. This functional form was chosen

since the $v_{\lambda\mu}$ were nearly linear on a semilogarithmic plot. For the rare cases where extrapolation to smaller R was needed, it was done using the fit for the $R=3\text{--}4$ interval. The one place where this fitting procedure could not be used due to a sign change was for v_{40} , $R=3\text{--}4$; since this component of the potential is very small, it was fixed by setting $v_{40}(R=3) = -1.0\text{ cm}^{-1}$. Finally, where the potential $v_{\lambda\mu}(R=8)$ was indicated as less than 0.1 cm^{-1} it was set arbitrarily as $v_{\lambda\mu}(R=8) = 0.1\text{ cm}^{-1}$ for purposes of providing a fit. Extrapolation at larger R was done with the fit to the $R=7\text{--}8$ interval, giving rapid exponential decay of the interaction.

Using this potential CS/B6 scattering calculations as described above were performed at a collision energy of 200 cm^{-1} , giving a cross section for the (1,1) doublet of $9.5\text{ \AA}^2$. To check the accuracy of the CS scattering approximation, close coupling calculations were done with the same basis (CC/B6) giving a value of $9.2\text{ \AA}^2$. To consider convergence with basis set, a calculation using the ten levels with $j=1, 2$, and 3 (CS/B10) was performed giving a value of $9.4\text{ \AA}^2$. It is apparent that, while the HF potential is somewhat better than the EG model, it still predicts only half the observed pressure broadening cross section.

C. Modifications of the HF potential

The HF method is expected to be accurate for the short-range repulsive part of the potential. It does not, however, account for the dispersion energy which dominates the long-range attraction, and, in fact, does not predict a van der Waals minimum at all. In principle, the HF method correctly predicts the long-range induction forces such as the dipole-induced dipole and the quadrupole-induced dipole. However, because they are a very small component of the total energy at large R these terms are often lost in the numerical error; the potential of Davis *et al.*¹⁰ does not appear to have the asymptotic behavior expected from the usual multipole perturbation expansion. It seems reasonable to attempt to improve on the HF potential by adding long-range dispersion and induction forces without significantly altering the short-range repulsive part.

It is noteworthy in this context that Anderson¹¹ was able to account satisfactorily for $\text{NH}_3\text{-He}$ pressure broadening using only a quadrupole-induced-dipole term for the potential. However, that calculation can be criticized in several points. It had to employ an arbitrary cutoff procedure at short interaction distances where the induction forces become infinite. It also assumed that the pressure broadening cross section was simply the intradoublet inelastic cross section. Finally, it treated dynamics within a semiclassical second-order perturbation framework. All of these approximations can be shown to be questionable. Nonetheless, the results suggest that long-range forces may be important to describe $\text{NH}_3\text{-He}$ pressure broadening.

Several recent studies have suggested that intermolecular forces might be well represented as the sum of a short-range exponential repulsion and a long-range attractive multipole dispersion series,

$$V(R) = A e^{-BR} - C_6 R^{-6} - C_8 R^{-8} - \dots, \quad (3)$$

where the exponential repulsion is generally taken from a HF calculation.¹²

Accordingly we have considered modifying the spherical part (v_{00}) of the HF potential by adding the leading dispersion term $-C_6/R^6$. The value of C_6 was adjusted in an attempt to mimic the Lennard-Jones fit to beam scattering data. This led ultimately to the choice of two values, $C_{6,\text{small}} = 8.56 \times 10^6 \text{ cm}^{-1} \text{ bohr}^6$ and $C_{6,\text{large}} = 1.22 \times 10^7 \text{ cm}^{-1} \text{ bohr}^6$. The values for the position of zero energy σ , minimum energy r_m , and well depth ϵ , for these choices can be compared with values inferred from beam data:

	σ , bohr	r_m , bohr	$(4\pi)^{1/2} \epsilon$, cm ⁻¹
$C_{6,\text{small}}$	6.30	7.15	48
$C_{6,\text{large}}$	6.05	6.75	80
Experimental	5.84	6.55	64

It is apparent that a single value of C_6 cannot be found which agrees with both the position and depth of the Lennard-Jones well inferred from experiment. It is not clear how much weight to place on the experimental parameters since they are strongly influenced by the assumed form of the interaction, and values between $C_{6,\text{small}}$ and $C_{6,\text{large}}$ give reasonable accord. It should be noted that the R^{-6} term has not been cut off or damped at small R . With these values for C_6 it makes relatively small changes to the energy on the repulsive wall ($R \sim 3\text{--}5$ bohr).

To account for the quadrupole-induced dipole term, we have followed a similar procedure and added the asymptotic form to the potential at all values of R . This term can be written as

$$\begin{aligned} V_{\text{qid}} &= -6\alpha\mu Q \cos^3\theta / R^7 \\ &= -6\alpha\mu Q R^{-7} (1/5) \{ (12\pi)^{1/2} Y_{10} + 2(4\pi/7)^{1/2} Y_{30} \}, \end{aligned} \quad (4)$$

where α is the polarizability of He and μ and Q are the dipole and quadrupole moments of NH₃. Unfortunately, the value for the quadrupole moment of NH₃ is not well known. (See, e.g., Ref. 13.) Anderson¹¹ suggested a point charge model which gave a value $Q = -1.325$ buckingham (1 buckingham = $10^{-26} \text{ esu cm}^2$) which, as discussed above, when used in conjunction with his model for pressure broadening gave good agreement with experiment. Early theoretical calculations suggested a smaller value, on the order of -0.3 buckingham; but a more recent *ab initio* calculation¹⁴ gave a larger value of about -3.0 buckingham. In order to test the sensitivity of results to choice of quadrupole moment, we have considered two values, the one chosen by Anderson, which seems a reasonable compromise, and a much larger value, $Q = -9.1$ buckingham. The actual potential terms due to these are given by

Q	v_{10}	v_{30}
-1.325	$-1.23 \times 10^6 / R^7$	$-0.54 \times 10^6 / R^7$
-9.1	$-8.45 \times 10^6 / R^7$	$-3.71 \times 10^6 / R^7$

TABLE I. Pressure broadening cross sections in Å² for the (1,1) NH₃ doublet computed from the Hartree-Fock potential augmented by various combinations of assumed long-range dispersion (C_6) and quadrupole-induced-dipole interactions.^a

	Q , buckingham		
C_6	0.0	-1.325	-9.1
none	9.4	10.4	17.3
$C_{6,\text{small}}$	16.7	18.1	26.4
$C_{6,\text{large}}$	20.8	22.2	30.6

^aSee text for further details of the potential.

where the energy is in cm⁻¹ and distances are in bohr. Again, these terms were not cut off at small R as they made only a small contribution in the accessible region of the repulsive interaction.

D. Pressure broadening with the modified HF potential

Using the HF potential plus various combinations of the long-range dispersion (C_6) interaction and quadrupole-induced dipole (V_{qid}) interaction, pressure broadening cross sections for the (1,1) doublet were obtained from CS/B10 scattering calculations at a collision energy of 200 cm⁻¹. Results are summarized in Table I. (A few additional calculations were done which suggested that the cross section is relatively insensitive to spectral level. For the potential with $C_{6,\text{small}}$ and $Q = -1.325$ the cross sections for the (2,2) and (3,3) doublets were found to be 19.5 and 19.3 Å², respectively, which can be compared with the value for the (1,1) doublet of 18.1 Å². Further, dependence on collision energy was also found to be small.)

These results are quite interesting, especially when compared with the experimental value of 18 Å². First, it appears that the quadrupole-induced dipole term used by itself in conjunction with the short-range repulsive HF interaction cannot adequately explain the observed pressure broadening. Even the unrealistically large value for the quadrupole moment of $Q = -9.1$ buckingham gives results just barely consistent with experiment. Second, it appears, on the other hand, that addition of a long-range attraction to only the spherical part of the HF potential does give good agreement with experiment. (Recall that the range of C_6 values used here seems to be realistic by comparison with beam scattering data.) This result might at first seem surprising since pressure broadening is associated with inelastic transitions and these are not induced by the isotropic part of the potential. The physical basis, however, is straightforward. Addition of a long-range attractive part to the spherical potential allows the helium to approach much closer to the NH₃ during a collision, causing it to exert a stronger torque due to the larger anisotropy of the potential at small distances. Finally, it is worth noting that, as for the EG potential, the pressure broadening cross section is only some 3%–5% larger than the total inelastic cross section, and the intradoublet cross section is only a fraction, typically one-third to one-half, of this value.

Consideration of the leading terms in the asymptotic multipole series suggests the possible importance of two other interactions, the dipole-induced dipole which contributes to v_{00} and v_{20} , and the anisotropy in the dispersion due to the polarizability anisotropy of NH₃ which contributes to v_{20} . A simple estimate indicates that the dipole-induced dipole can be neglected compared with the dispersion terms that have the same distance and angle dependence. The polarizability anisotropy of NH₃ is $3(\alpha_{\parallel} - \alpha_{\perp})/(\alpha_{\parallel} + 2\alpha_{\perp}) \approx 0.1$. Calculations were done adding this term to the potential along with $C_{6,\text{small}}$ and V_{qld} with $Q = -1.325$; this increased the (1,1) pressure broadening cross section from 18.1 to 19.8 Å². (After completing these calculations it was discovered that the actual value used, $v_{20} = -1.36 \times 10^6/R^6$, was erroneously computed as $(4\pi)^{1/2}$ larger than the value appropriate for the $C_{6,\text{small}} v_{00}$ term. As shown in more detail in the next section, addition of this term caused relatively small changes in cross sections and it was not deemed worthwhile to repeat all the affected calculations.)

It is clear that this final potential is consistent with pressure broadening data. Furthermore, it is consistent with HF calculations for the short-range repulsive interaction and with estimates for the leading terms in the long-range multipole expansion. The isotropic part is in reasonable accord with beam scattering data. However, it is apparent that modest adjustments of the potential parameters—especially those which describe the region of the van der Waals minimum—could be made which would still provide comparably good agreement with available information.

E. State-to-state cross sections and rate constants

Since an important goal of this study is to provide state-to-state rate constants needed to analyze radio-astronomical observations, it is useful to consider the sensitivity of specific cross sections to uncertainties in the assumed potential. To this end scattering calculations have been performed for five of the potentials discussed above. All of them consisted of the Hartree-Fock potential plus differing long-range attractive parts. Potential 1 includes the $C_{6,\text{small}}$ dispersion term, the quadrupole-induced dipole with $Q = -1.325$, and the polarizability anisotropy term. Potential 2 is potential 1 minus the polarizability anisotropy term. Potentials 3 and 4 include only the isotropic dispersion term, the former using $C_{6,\text{small}}$ and the latter $C_{6,\text{large}}$. Finally, potential 5 is the HF short-range potential with no long-range terms. Table II presents state-to-state cross sections at a total energy of 120.445 cm⁻¹ obtained from CS/B10 calculations. For potential 1 a CS/B16 calculation (which includes in addition the $j=4$ levels) was also done to check convergence of cross sections with basis set, with the conclusion that values here are converged to better than 10%–15%. To test the adequacy of the CS approximation additional CC/B10 calculations were done for potential 1 with the conclusion that CS is accurate, in general, to better than 10%, although errors of up to 30%–50% are occasionally encountered for small cross sections. It is seen from Table II that individual cross sections are sensitive to variations in the potential; however, the larger cross sections (which are more im-

TABLE II. State-to-state cross sections in Å² for NH₃-He at a total energy of 120.445 cm⁻¹ for several potential energy surfaces described in the text.

initial	final	pot'l 1	pot'l 2	pot'l 3	pot'l 4	pot'l 5
1 1 l →	1 1 u	9.15	7.79	6.59	8.92	3.05
	2 1 l	0.74	0.05	0.05	0.12	0.03
	2 1 u	4.31	4.15	3.71	5.13	1.52
	2 2 l	2.08	2.87	2.92	4.01	1.16
	2 2 u	0.05	0.03	0.03	0.05	0.01
	3 1 l	0.04	0.02	0.01	0.03	0.0
	3 1 u	0.18	0.13	0.14	0.22	0.01
	3 2 l	0.14	0.18	0.17	0.31	0.03
2 1 l →	3 2 u	0.24	0.27	0.26	0.66	0.02
	2 1 u	5.28	4.40	4.10	6.14	1.39
	2 2 l	1.92	2.17	2.20	3.23	0.75
	2 2 u	0.74	0.81	0.77	1.59	0.14
	3 1 l	0.07	0.04	0.03	0.08	0.0
	3 1 u	0.77	0.81	0.74	1.25	0.05
2 2 l →	3 2 l	0.83	0.83	0.83	1.57	0.12
	3 2 u	0.12	0.11	0.10	0.23	0.01
	2 2 u	12.19	10.71	9.01	12.63	3.66
	3 1 l	0.14	0.14	0.14	0.25	0.01
	3 1 u	0.14	0.15	0.14	0.36	0.0
3 2 l →	3 2 l	0.19	0.04	0.04	0.11	0.01
	3 2 u	1.04	1.04	1.00	1.70	0.18

portant) do not vary all that much, rarely by as much as a factor of two.

Potential 1 was chosen as the “best” estimate for the true interaction, and a series of calculations were done with this potential to obtain cross sections as a function of energy. Since NH₃ is observed in many interstellar sources with low temperatures (on the order of 10–20 K) special attention was paid to choosing energies near threshold. In particular, CS/B10 calculations were done for 18 energies in the range 20–175 cm⁻¹. In addition CS/B16 and CS/B24 calculations were done for an additional nine energies to 400 cm⁻¹. Also, in these calculations the energy splittings in the vibration-rotation doublets were taken into account by lowering and raising the appropriate levels by 0.4 cm⁻¹ from rigid rotor values. [In all previous calculations, including those discussed above, these energy levels were assumed to be degenerate so that the relations $\sigma(jkl-j'k'l) = \sigma(jku-j'k'u)$ and $\sigma(jkl-j'k'u) = \sigma(jku-j'k'l)$ are rigorously true.] Of course, these energy splittings cause very small changes except at the lowest collision energies.

The computed collision cross sections were integrated numerically over Boltzmann distributions for a number of temperatures to 300 K. Some caveats must be given for the rates above 100 K as these required rather extensive (linear) extrapolation of the cross sections as a function of collision energy. The resulting rate constants are listed in Table III.

F. Comparison with double resonance experiments

Using the state-to-state rates at 300 K, the first-order (i.e., single collision) changes in intensity for the double resonance signals have been calculated using Eq. (1). Although there is some question about absolute values (due, e.g., to the possible exclusion of processes involving higher levels which contribute to the total rate in the denominator and to the definition of k_{tot}

TABLE III. Rate constants as a function of kinetic temperature, for excitation in $\text{NH}_3\text{-He}$ collisions. Units are $\text{cm}^3 \text{sec}^{-1}$; value in parenthesis is power of ten. De-excitation rates can be obtained from detailed balance.

		Temperature, K			
Initial	Final	15	30	100	300
1, 1l	1, 1u	6.7(-11)	6.8(-11)	7.8(-11)	8.7(-11)
	2, 2l	9.1(-13)	4.1(-12)	1.5(-11)	2.7(-11)
	2, 2u	6.8(-14)	2.2(-13)	2.5(-13)	4.1(-13)
	2, 1l	1.2(-13)	9.6(-13)	5.0(-12)	1.1(-11)
	2, 1u	7.0(-13)	5.4(-12)	2.9(-11)	6.2(-11)
	3, 2l	6.6(-16)	5.8(-14)	1.8(-12)	5.8(-12)
	3, 2u	1.0(-15)	9.6(-14)	3.9(-12)	1.7(-11)
	3, 1l	1.4(-16)	1.6(-14)	5.1(-13)	2.4(-12)
	3, 1u	5.1(-16)	6.7(-14)	2.7(-12)	1.0(-11)
	4, 4l	8.7(-18)	1.0(-14)	1.8(-12)	1.1(-11)
	4, 4u	8.1(-19)	1.1(-15)	3.2(-13)	2.6(-12)
	4, 2l	3.8(-20)	1.9(-16)	1.3(-13)	1.3(-12)
	4, 2u	1.3(-19)	7.0(-16)	4.7(-13)	4.3(-12)
	4, 1l	6.3(-21)	5.2(-17)	4.6(-14)	4.7(-13)
	4, 1u	3.1(-20)	2.5(-16)	2.0(-13)	1.8(-12)
1, 1u	2, 2l	8.9(-14)	2.5(-13)	3.8(-13)	4.6(-13)
	2, 2u	9.0(-13)	4.0(-12)	1.5(-11)	2.7(-11)
	2, 1l	8.0(-13)	5.8(-12)	3.0(-11)	6.3(-11)
	2, 1u	1.3(-13)	9.5(-13)	4.9(-12)	1.1(-11)
	3, 2l	1.2(-15)	1.1(-13)	4.1(-12)	1.8(-11)
	3, 2u	6.7(-16)	5.7(-14)	1.8(-12)	6.0(-12)
	3, 1l	5.8(-16)	7.3(-14)	2.8(-12)	1.1(-11)
	3, 1u	1.4(-16)	1.6(-14)	5.3(-13)	2.7(-12)
	4, 4l	8.8(-19)	1.1(-15)	3.2(-13)	2.7(-12)
	4, 4u	9.1(-18)	1.0(-14)	1.8(-12)	1.1(-11)
	4, 2l	1.5(-19)	7.6(-16)	4.9(-13)	4.4(-12)
	4, 2u	3.8(-20)	1.9(-16)	1.3(-13)	1.3(-12)
	4, 1l	3.5(-20)	2.8(-16)	2.1(-13)	1.8(-12)
	4, 1u	6.7(-21)	5.4(-17)	5.5(-14)	6.3(-13)
2, 2l	2, 2u	7.9(-11)	8.1(-11)	9.5(-11)	1.2(-10)
	2, 1l	3.9(-12)	6.9(-12)	1.3(-11)	1.7(-11)
	2, 1u	1.0(-12)	2.1(-12)	6.1(-12)	1.4(-11)
	3, 2l	9.9(-15)	2.2(-13)	2.6(-12)	7.4(-12)
	3, 2u	5.3(-14)	1.1(-12)	1.5(-11)	3.9(-11)
	3, 1l	4.6(-15)	1.5(-13)	2.4(-12)	6.6(-12)
	3, 1u	4.7(-15)	1.7(-13)	3.6(-12)	1.3(-11)
	4, 4l	3.2(-17)	9.5(-15)	7.2(-13)	3.5(-12)
	4, 4u	2.9(-18)	7.8(-16)	4.1(-14)	1.6(-13)
	4, 2l	8.7(-19)	1.1(-15)	2.2(-13)	1.3(-12)
	4, 2u	2.8(-18)	3.5(-15)	7.4(-13)	4.5(-12)
	4, 1l	2.0(-19)	4.3(-16)	1.5(-13)	1.2(-12)
	4, 1u	7.4(-19)	1.6(-15)	5.1(-13)	3.7(-12)
2, 2u	2, 1l	1.1(-12)	2.2(-12)	6.2(-12)	1.4(-11)
	2, 1u	4.0(-12)	7.0(-12)	1.3(-11)	1.7(-11)
	3, 2l	6.1(-14)	1.2(-12)	1.5(-11)	4.0(-11)
	3, 2u	1.0(-14)	2.2(-13)	2.5(-12)	7.4(-12)
	3, 1l	5.3(-15)	1.8(-13)	3.7(-12)	1.3(-11)
	3, 1u	4.8(-15)	1.6(-13)	2.4(-12)	6.7(-12)
	4, 4l	3.4(-18)	8.6(-16)	4.4(-14)	1.8(-13)
	4, 4u	3.3(-17)	9.6(-15)	7.2(-13)	3.5(-12)
	4, 2l	3.3(-18)	3.8(-15)	7.8(-13)	4.6(-12)
	4, 2u	9.0(-19)	1.1(-15)	2.2(-13)	1.3(-12)
	4, 1l	8.5(-19)	1.7(-15)	5.3(-13)	3.8(-12)
	4, 1u	2.1(-19)	4.3(-16)	1.5(-13)	1.1(-12)
2, 1l	2, 1u	2.8(-11)	3.0(-11)	3.9(-11)	4.7(-11)
	3, 2l	1.1(-13)	1.3(-12)	9.5(-12)	1.9(-11)
	3, 2u	1.6(-14)	1.8(-13)	9.9(-13)	2.0(-12)
	3, 1l	6.0(-15)	1.2(-13)	1.3(-12)	3.9(-12)
	3, 1u	6.1(-14)	1.3(-12)	1.6(-11)	4.8(-11)
	4, 4l	5.8(-16)	9.5(-14)	3.7(-12)	1.2(-11)
	4, 4u	2.8(-16)	4.7(-14)	2.3(-12)	9.8(-12)
	4, 2l	6.2(-18)	4.5(-15)	6.4(-13)	3.5(-12)
	4, 2u	9.2(-18)	7.1(-15)	1.2(-12)	7.1(-12)
	4, 1l	1.0(-18)	8.9(-16)	1.3(-13)	6.9(-13)
	4, 1u	2.0(-18)	2.0(-15)	3.6(-13)	2.3(-12)

TABLE III (Continued)

		Temperature, K			
Initial	Final	15	30	100	300
2, 1u	3, 2l	2.5(-14)	2.3(-13)	1.1(-12)	2.0(-12)
	3, 2u	1.1(-13)	1.3(-12)	9.5(-12)	2.0(-11)
	3, 1l	7.0(-14)	1.4(-12)	1.7(-11)	4.9(-11)
	3, 1u	6.0(-15)	1.2(-13)	1.3(-12)	4.0(-12)
	4, 4l	3.1(-16)	5.0(-14)	2.3(-12)	1.0(-11)
	4, 4u	6.1(-16)	9.6(-14)	3.7(-12)	1.2(-11)
	4, 2l	1.1(-17)	7.8(-15)	1.2(-12)	7.3(-12)
	4, 2u	6.5(-18)	4.6(-15)	6.5(-13)	3.6(-12)
	4, 1l	2.9(-18)	2.5(-15)	3.8(-13)	2.3(-12)
	4, 1u	1.6(-18)	1.2(-15)	1.5(-13)	7.7(-13)
3, 2l	3, 2u	2.5(-11)	3.3(-11)	4.9(-11)	6.7(-11)
	3, 1l	4.4(-12)	8.1(-12)	1.5(-11)	2.0(-11)
	3, 1u	4.3(-13)	8.9(-13)	2.4(-12)	4.9(-12)
	4, 4l	6.2(-15)	1.0(-13)	9.6(-13)	2.6(-12)
	4, 4u	2.1(-15)	3.8(-14)	5.2(-13)	1.8(-12)
	4, 2l	5.6(-16)	3.8(-14)	1.0(-12)	3.7(-12)
	4, 2u	4.0(-15)	2.7(-13)	8.0(-12)	3.0(-11)
	4, 1l	5.2(-16)	5.8(-14)	2.0(-12)	7.2(-12)
	4, 1u	3.5(-16)	4.2(-14)	1.8(-12)	7.5(-12)
3, 2u	3, 1l	5.1(-13)	9.7(-13)	2.5(-12)	5.1(-12)
	3, 1u	4.6(-12)	8.2(-12)	1.5(-11)	2.1(-11)
	4, 4l	2.3(-15)	4.0(-14)	5.3(-13)	1.8(-12)
	4, 4u	6.4(-15)	1.0(-13)	9.7(-13)	2.7(-12)
	4, 2l	4.7(-15)	3.0(-13)	8.3(-12)	3.1(-11)
	4, 2u	5.5(-16)	3.7(-14)	1.0(-12)	3.7(-12)
	4, 1l	4.1(-16)	4.6(-14)	1.9(-12)	7.6(-12)
	4, 1u	5.4(-16)	5.8(-14)	2.1(-12)	7.6(-12)
3, 1l	3, 1u	1.1(-11)	1.2(-11)	1.8(-11)	2.6(-11)
	4, 4l	1.8(-13)	1.6(-12)	7.2(-12)	1.2(-11)
	4, 4u	1.4(-13)	1.3(-12)	6.9(-12)	1.5(-11)
	4, 2l	8.7(-15)	3.4(-13)	5.9(-12)	1.7(-11)
	4, 2u	1.2(-15)	4.9(-14)	9.2(-13)	2.9(-12)
	4, 1l	3.3(-16)	2.2(-14)	5.6(-13)	2.1(-12)
	4, 1u	5.3(-16)	3.6(-13)	9.9(-12)	3.7(-11)
3, 1u	4, 4l	1.5(-13)	1.3(-12)	7.0(-12)	1.5(-11)
	4, 4u	1.9(-13)	1.6(-12)	7.2(-12)	1.2(-11)
	4, 2l	1.5(-15)	5.6(-14)	9.7(-13)	3.0(-12)
	4, 2u	8.9(-15)	3.5(-13)	5.9(-12)	1.7(-11)
	4, 1l	6.1(-15)	4.0(-13)	1.0(-11)	3.8(-11)
	4, 1u	3.5(-16)	2.2(-14)	5.6(-13)	2.0(-12)
4, 4l	4, 4u	2.1(-11)	5.3(-11)	1.0(-10)	1.5(-10)
	4, 2l	2.7(-15)	4.7(-14)	4.9(-13)	1.4(-12)
	4, 2u	2.2(-15)	4.0(-14)	5.2(-13)	1.7(-12)
	4, 1l	8.7(-15)	1.7(-13)	1.9(-12)	5.2(-12)
	4, 1u	1.0(-14)	2.1(-13)	2.7(-12)	8.2(-12)
4, 4u	4, 2l	2.2(-15)	4.0(-14)	5.3(-13)	1.7(-12)
	4, 2u	2.6(-15)	4.5(-14)	4.9(-13)	1.4(-12)
	4, 1l	1.1(-14)	2.2(-13)	2.7(-12)	8.3(-12)
	4, 1u	8.8(-15)	1.7(-13)	1.9(-12)	5.2(-12)
4, 2l	4, 2u	8.3(-12)	1.7(-11)	3.1(-11)	4.6(-11)
	4, 1l	3.0(-12)	7.3(-12)	1.6(-11)	2.6(-11)
	4, 1u	1.6(-13)	3.9(-13)	1.1(-12)	2.2(-12)
4, 2u	4, 1l	1.8(-13)	4.2(-13)	1.1(-12)	2.3(-12)
	4, 1u	3.1(-12)	7.3(-12)	1.6(-11)	2.6(-11)
4, 1l	4, 1u	4.6(-12)	6.6(-12)	1.0(-11)	1.4(-11)

as discussed in Sec. II B) the sign of the result depends only on the difference in lower-to-lower and lower-to-upper rates between the pump and signal levels. Results are given in Table IV where they are compared with experimental values and with values obtained using the EG potential.⁴ The most striking conclusion is that the present potential (HF plus long-range) gives predictions in excellent overall accord with those from the EG potential. Agreement with experiment is mixed. For

TABLE IV. Comparison of theoretical and experimental values for $\text{NH}_3\text{-He}$ double resonance changes in intensity (%), from the first-order expression, Eq. (1), using state-to-state rates for two assumed potentials, the Hartree-Fock plus long range described in the text and an electron gas model.

$\Delta k = 3$					$\Delta k = 0$				
signal	pump	HF + LR	EG ^a	expt. ^a	signal	pump	HF + LR	EG ^a	expt. ^a
1,1	2,2	-12	-13	-10.0	1,1	2,1	+22	+5	-3.5
1,1	3,2	+5	+0.5	0.0	1,1	3,1	+3	+2	-2.8
1,1	4,4	-4	-11	-9.4	1,1	4,1	+0.6	+1.5	-0.3
1,1	4,2	+1	+1	+0.2	2,2	3,2	+14	+5.5	-4.2
2,2	2,1	-1	-4	-3.0	2,2	4,2	+1	+2	0.0
2,2	3,1	+3	+0.2	-0.3	2,1	3,1	+19	+6	+2.5
2,2	4,1	+1	+0.7	0.0	2,1	4,1	+0.7	+0.2	-2.2
2,1	3,2	-7	-7	-5.0	3,2	4,2	+12	+5	0.0
2,1	4,4	-1	-7	-6.0	3,1	4,1	+15	+8	+4.4
2,1	4,2	+2	+2	0.0					
3,2	3,1	-7	-6	-2.5					
3,2	4,1	0	-1	0.0					
3,1	4,4	+1	-3	-0.6					
3,1	4,2	-6	-6	-5.0					
4,4	3,2	-0.4	...	-1.0					
4,4	4,2	+0.1	...	0.0					
4,4	4,1	+1	...	0.0					
4,2	4,1	-11	...	-5.0					

^aEG values from Davis and Boggs, Ref. 4; experimental values from Oka, Refs. 2 and 3 or as quoted in Ref. 4.

the $\Delta k = 0$ cases the agreement with experiment is poor; both potentials predict consistently positive $\Delta I/I$ while experiment finds mostly negative values. For the $\Delta k = 3$ cases, on the other hand, agreement between theory and experiment is good. In both cases the EG results are perhaps in marginally better accord with experiment, but the difference is not significant.

Although it is conceivable that the remaining discrepancies between theory and experiment are due to inaccuracies in the potential, the present results make this seem unlikely. For example, none of the variations in the long-range form of the potential which were considered above (cf. Table II) would reverse the sign of the theoretical value in cases where it disagrees with experiment. The extrapolation of cross sections to higher energies used here is also not likely to be a source of error, since only small temperature dependence is observed. Further progress toward understanding these results will probably require consideration of the dependence of collision rates on magnetic quantum number, including a proper analysis of this aspect of the experimental observations. In this context it should be noted that there is some question about the validity of the CS scattering approximation for Δm transitions, so that more expensive CC calculations may be required.

V. CONCLUSIONS

Collision cross sections have now been computed for $\text{NH}_3\text{-He}$ collisions at thermal and subthermal energies using dynamical approximations of documented accuracy and several different intermolecular potentials. The potentials were obtained from (1) a Gordon-Kim⁹ electron gas model, (2) Hartree-Fock calculations,¹⁰ and (3) Hartree-Fock calculations augmented semiempirically with long-range dispersion and induction terms. Re-

sults have been compared with the experimental data available for this system: beam scattering, pressure broadening, and microwave double resonance. The beam scattering experiments provide information about the isotropic interaction such as the position and depth of the van der Waals minimum. Pressure broadening gives a measure of the *total* inelasticity. And microwave double resonance gives information about the *relative* size of different inelastic cross sections.

It appears that the electron gas model is a relatively poor approximation for this system. In particular the van der Waals well depth is predicted to be about three times larger than values inferred from beam scattering data. Also, the pressure broadening cross section predicted from this potential is more than four times smaller than experimental values; i.e., the EG potential predicts too little inelasticity in $\text{NH}_3\text{-He}$ collisions. Its success in predicting the relative size of different cross sections, as judged by comparison with double resonance experiments, is mixed.

The Hartree-Fock method gives only the short-range repulsive interaction. This potential by itself also predicts too little inelasticity when compared with pressure broadening experiments. By adding long-range dispersion interactions to the HF potential, however, it is possible to obtain a van der Waals minimum in accord with beam scattering data. Using reasonable values for the long-range dispersion and quadrupole-induced dipole induction in addition to the HF short-range repulsion also gives good agreement with the total inelasticity as measured by pressure broadening. As with the EG potential, however, the success of the Hartree-Fock plus long-range interaction in predicting the relative size of different cross sections, as judged by comparison with double resonance experiments, is mixed. Further,

none of the variations in the long-range parts of the potential which were considered in Sec. IV C improve this agreement.

The remaining discrepancies between predicted and observed double resonance signals are the major obstacle to a satisfactory understanding of excitation in NH_3 -He collisions. It therefore seems important in concluding to consider the likely sources of these discrepancies and to suggest methods for resolving them. As discussed above, and in Ref. 4, only two possibilities seem likely, (1) inaccuracies in the potential surface and (2) improper treatment of the dependence of collision rates on magnetic quantum number.

The major criticism of the interaction potential developed here is the rather *ad hoc* procedure used to join the asymptotic long-range potential with the short-range Hartree-Fock interaction. It is certainly possible to improve the description of the potential in the region of the minimum and the low energy part of the repulsive interaction. This could be done by using more sophisticated procedures for joining long-range and short-range forces, for example including higher order dispersion terms; such possibilities are discussed in detail in Ref. 12. Alternatively, the correlation energy could be obtained directly from *ab initio* calculations, for example, using configuration interaction wave functions (cf. the study of H_2CO -He by Garrison *et al.*¹⁵). While it is conceivable that an improved potential would correct the discrepancies with the double resonance data, as noted above this seems unlikely. The reasoning behind this statement is as follows. The relative sizes of various cross sections for this system appear to depend on the basic shape of the short-range repulsion. As discussed in Sec. IV. F, the double resonance predictions were not significantly altered by the various modifications to the long-range potential which were considered here. Further, the electron gas potential, although it differs in detail from the Hartree-Fock interaction, still has the same qualitative short-range shape; and it is noteworthy that these two potentials predict very similar double resonance behavior.

It therefore seems that the most likely problem is the way in which the dependence of cross sections on magnetic quantum numbers has been treated. The present calculations (and previous calculations) have considered only degeneracy-averaged cross sections, i.e., summed over final magnetic quantum number and averaged over initial quantum number. Further, the analysis of experimental results which leads to Eq. (1) assumes cross sections which are independent of magnetic quantum number. As noted in Refs. 3 and 4 the experiment does not, in fact, treat different magnetic quantum levels on an equal footing. It appears necessary to reconsider the experimental analysis, explicitly including the dependence on magnetic sublevels. In order to compare with theory it will then be necessary to calculate the *m*-dependence of collision cross sections.

It is reasonable to believe that the Hartree-Fock plus long-range interaction developed here provides an adequate description of the degeneracy averaged rates needed in radioastronomical studies. The interaction

potential is consistent with beam scattering data, and the total inelastic cross section is in good accord with pressure broadening measurements. It is hoped that proper consideration of magnetic sublevels will remove the remaining discrepancies with double resonance data; it should, perhaps, be noted that the overall agreement with experiment here is not that bad, with serious discrepancies in only a fraction of the cases. Further improvements of the potential in the region of the van der Waals minimum will, of course, cause some changes in individual rates; an estimate of the amount of uncertainty in the present values can be obtained by considering the changes in cross section found for reasonable variations in the long-range potential which are given in Table II.

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APPENDIX

The specific form and symmetry of the rotation-vibration wavefunctions for NH_3 have received much attention. Nonetheless, there still appears to be some confusion on this point, and there are errors in standard microwave spectroscopy texts including the one by Sugden and Kenney¹⁶ on which Davis and Boggs⁴ based their work and the frequently used text by Townes and Schawlow.¹⁷ Also, statements concerning parity made in previous papers by the author⁸ are incorrect. An attempt will be made here to review this subject and correct these errors. The effect of symmetry, including nuclear spin, and of vibrational (inversion) motion on scattering matrix elements will be treated in some detail.

It is necessary first to define coordinate systems. As always, there will be a space fixed coordinate (X, Y, Z). In order to specify a molecule fixed coordinate (x, y, z) it is necessary to label the three hydrogen atoms: $\text{H}_1, \text{H}_2, \text{H}_3$. Then the molecule fixed z -axis is chosen as the threefold symmetry axis, with positive direction defined in a right-handed sense by $\text{H}_1\text{H}_2\text{H}_3$. As noted previously,⁸ the molecule fixed coordinates which are used to define the intermolecular potential *must* correspond to the principal moment of inertia axes used to define the rotational wave function. Since the scattering formalism has used the fact that the xz plane is a plane of symmetry, it is necessary to place H_1 in this plane. This differs from the discussion in Refs. 16 and 17, both of which place H_1 in the yz plane, and this will prove to be a nontrivial distinction in forming properly symmetrized wave functions. The vibrational motion is described by a coordinate h , measured along the z -axis such that positive h implies that the nitrogen

is on the positive z -axis. The vibrational motion is sometimes called inversion, but this term will not be used here to avoid confusion with the symmetry operation of space inversion E^* , which corresponds to $x \rightarrow -x$, $y \rightarrow -y$, and $z \rightarrow -z$, and which is not the same as the vibrational inversion of the molecule between the two equilibrium values $h = \pm h_e$.

The rotational motion of the molecule is then described by the Euler angles θ, ϕ, χ which rotate (X, Y, Z) to (x, y, z) . Various conventions for the Euler angles exist. We will take θ and ϕ as the usual polar coordinates of the molecule fixed z -axis with respect to the space fixed axes, i.e., θ is the angle between z and Z , and ϕ is the angle from the X -axis to the projection of z on the XY plane. The angle χ represents rotation about the molecule fixed z -axis and is measured from the line of nodes (the intersection of the XY and xy planes) to the x -axis in the positive (i.e., $\text{H}_1\text{H}_2\text{H}_3$) direction.

One can then define "primitive" rotational functions $|JKM\rangle$ proportional to the rotational matrices; in particular

$$|JKM\rangle \sim e^{iM\phi} e^{iK\chi} d_{KM}^J(\theta), \quad (\text{A1})$$

where the reduced rotation matrix, d_{KM}^J is defined as in Edmonds.¹⁸ It is worth noting at this point that different conventions for the Euler angles and for the form of the rotational functions, as long as they are applied consistently, will generally introduce only ignorable overall phase factors and will not change the symmetry arguments developed below. The one point where particular care must be exercised, however, is in the angle χ . This is because the final wave functions for symmetric (and asymmetric) tops will involve linear combinations of different K values, and changes in χ can introduce *relative* phase shifts in different terms, in addition to ignorable overall phase factors.

The vibrational wave functions which depend on h will be written as $|+\rangle$ for the lower level and $|-\rangle$ for the upper level indicating that these are, respectively, even and odd functions of h . That is,

$$|\pm\rangle \sim f(h_e - h) \pm f(h_e + h), \quad (\text{A2})$$

where $f(q)$ is some function whose only relevant property is that it is strongly peaked near $q = 0$.

Since the Hamiltonian is invariant to space inversion, E^* , acceptable wave functions must be eigenfunctions of E^* with eigenvalues of ± 1 . The effect of E^* on the rotational coordinates is discussed in Refs. 16 and 17. Since the order of H_1 , H_2 , and H_3 is not changed, the molecule fixed z -axis is unchanged so that $\theta \rightarrow \theta$ and $\phi \rightarrow \phi$. However, $\chi \rightarrow \chi + \pi$. Also, since the direction of the z -axis has not changed, but the molecule has been turned inside out, $h \rightarrow -h$. These coordinate changes are readily visualized by recalling that space inversion can be accomplished by rotating a molecule by π about any axis, followed by reflection in a plane perpendicular to that axis (in this case, rotation about the z -axis and reflection in the xy plane).

It is apparent from Eq. (A1) that these coordinate changes imply

$$E^* |JKM\rangle = e^{iK\pi} |JKM\rangle = (-)^K |JKM\rangle. \quad (\text{A3})$$

Thus, each primitive rotational function has the parity $(-)^K$ in contrast to the statement in Ref. 8 that these functions do not have definite parity. (The misconception in Ref. 8 arose from the analogy of symmetric top wave functions to Λ -doublets in linear molecules. However, the coordinate systems for these two cases are quite different. In the linear molecule the z -axis is defined by the two nuclei, and E^* takes $z \rightarrow -z$. This is in contrast to the discussion of the symmetric top given above where the direction of the z -axis is determined from the order of the three hydrogens and is unchanged by E^* .) It is apparent from Eq. (A2) that the vibrational functions have definite parity:

$$E^* |\pm\rangle = \pm |\pm\rangle. \quad (\text{A4})$$

The Hamiltonian is also invariant to interchange of hydrogen nuclei. Since these are fermions, interchange of any two—i.e., the symmetry operations (12), (13), (23)—must change the sign of the *total* wave function.

The effect on rotational coordinates of the operation (23) has been discussed in some detail in Ref. 16 and to a lesser extent in Ref. 17, both of which give the result that (23) changes the coordinates as:

$$\theta \rightarrow \pi - \theta, \quad \phi \rightarrow \pi + \phi, \quad \chi \rightarrow \pi - \chi, \quad h \rightarrow -h. \quad (\text{A5})$$

The effect of these coordinate changes on $|JKM\rangle$, however, is given incorrectly in both Refs. 16 and 17. Using the fact¹⁸ that

$$d_{KM}^J(\pi - \theta) = (-)^{J+M} d_{-KM}^J(\theta) \quad (\text{A6})$$

it is readily seen from Eq. (A1) that the correct transformation is

$$(23) |JKM\rangle = (-)^{J+K} |J-KM\rangle. \quad (\text{A7})$$

Unfortunately, Eqs. (A5) and (A7) are not relevant here. They apply only when H_1 is in the yz plane, the convention adopted in Refs. 16 and 17. As discussed above, to be consistent with the form of the intermolecular potential it is necessary to place H_1 in the xz plane. In that case it can be shown that (23) causes the following coordinate changes:

$$\theta \rightarrow \pi - \theta, \quad \phi \rightarrow \pi + \phi, \quad \chi \rightarrow -\chi, \quad h \rightarrow -h. \quad (\text{A8})$$

Then

$$(23) |JKM\rangle = (-)^J |J-KM\rangle \quad (\text{A9})$$

and

$$(23) |\pm\rangle = \pm |\pm\rangle. \quad (\text{A10})$$

It is apparent from Eq. (A9) that linear combinations of the degenerate functions $|JKM\rangle$ and $|J-KM\rangle$ will be necessary to obtain eigenfunctions of (23). This is further complicated if one considers interchange of other hydrogens, e.g., (12). It can be seen that (12) may be obtained as (23) followed by rotation by $2\pi/3$ about the z -axis, i.e.,

$$(12) = C_3 (23). \quad (\text{A11})$$

It is immediately obvious that C_3 affects only the angle χ , $\chi \rightarrow \chi + 2\pi/3$, so that

$$(12) |JKM\rangle = (-)^J e^{-2i\pi K/3} |J-KM\rangle. \quad (\text{A12})$$

By similar reasoning

$$(13) |JKM\rangle = (-)^J e^{-4i\pi K/3} |J-KM\rangle. \quad (A13)$$

Only if K is a multiple of 3 are the exponential factors in Eqs. (A12) and (A13) unity.

As discussed in some detail in Refs. 16 and 17, in order to obtain proper eigenfunctions of the permutation operators it is necessary to consider the nuclear spin functions as well as the rotation functions. In particular, one can define a wave function

$$|JKM\epsilon\rangle = 6^{-1/2} \{ |JKM\rangle (\beta\alpha\alpha + \alpha\beta\alpha e^{2i\pi K/3} + \alpha\alpha\beta e^{4i\pi K/3}) + \epsilon |J-KM\rangle (\beta\alpha\alpha + \alpha\beta\alpha e^{-2i\pi K/3} + \alpha\alpha\beta e^{-4i\pi K/3}) \} \quad (A14)$$

where $\epsilon = \pm 1$ and K is now restricted to non-negative integers. The spin functions $\gamma_1\gamma_2\gamma_3$ are the nuclear spins associated with H₁, H₂, and H₃; and α corresponds to $M_I = \frac{1}{2}$, β to $M_I = -\frac{1}{2}$. For $K = 3n$, the spin function in Eq. (A14) reduces to $(\beta\alpha\alpha + \alpha\beta\alpha + \alpha\alpha\beta)$ which, along with $\alpha\alpha\alpha$, $(\alpha\beta\beta + \beta\alpha\beta + \beta\beta\alpha)$, and $\beta\beta\beta$ gives a quartet ($I = 3/2$) corresponding to ortho-NH₃. For para-NH₃, $K = 3n \pm 1$, and the spin function in Eq. (14) averaged over rotational motion reduces to a doublet ($I = \frac{1}{2}$) function $(2\beta\alpha\alpha - \alpha\beta\alpha - \alpha\alpha\beta)$, the other member of which is $(2\alpha\beta\beta - \beta\alpha\beta - \beta\beta\alpha)$. Recalling that the permutation operations also affect the nuclear spin function—for example, $(12)\beta\alpha\alpha \rightarrow \alpha\beta\alpha$ —it can be seen by direct substitution of Eqs. (A9), (A12), and (A13) that

$$(12) |JKM\epsilon\rangle = (13) |JKM\epsilon\rangle = (23) |JKM\epsilon\rangle = \epsilon(-)^J |JKM\epsilon\rangle. \quad (A15)$$

The question of permissible values of ϵ will be deferred until after a consideration of the effect of nuclear spin functions on matrix elements.

Since neither the intermolecular potential nor the Hamiltonian depend on nuclear spin, it is clear that only terms which are diagonal in nuclear spin functions will survive in matrix elements. Thus,

$$\langle JKM\epsilon | V | J'K'M'\epsilon' \rangle = (1/6)(V_1 + \epsilon V_2 + \epsilon' V_3 + \epsilon\epsilon' V_4), \quad (A16)$$

where, for example,

$$\begin{aligned} V_1 &= \langle JKM | (\beta\alpha\alpha + \alpha\beta\alpha e^{-2i\pi K/3} + \alpha\alpha\beta e^{-4i\pi K/3}) \\ &\quad \times V(\beta\alpha\alpha + \alpha\beta\alpha e^{2i\pi K'/3} + \alpha\alpha\beta e^{4i\pi K'/3}) | J'K'M' \rangle \\ &= \langle JKM | V(1 + \exp[2i\pi(K' - K)/3] \\ &\quad + \exp[4i\pi(K' - K)/3]) | J'K'M' \rangle. \end{aligned} \quad (A17)$$

But the sum of three terms in parentheses in the last line vanishes identically unless $K' - K = 3n$, in which case each of the three terms is unity. The V_4 term has a similar constraint, and V_2 and V_3 vanish unless $K' + K = 3n$. However, these constraints are identical to the constraints due to the threefold axis of symmetry in the interaction potential. Thus, after integrating over spin coordinates one finds

$$\begin{aligned} \langle JKM\epsilon | V | J'K'M'\epsilon' \rangle &= \frac{1}{2} \langle JKM | + \epsilon \langle J-KM | \\ &\quad \times V(|J'K'M'\rangle + \epsilon' |J'-K'M'\rangle), \end{aligned} \quad (A18)$$

which is just the “parity symmetrized” matrix element discussed in Ref. 8. It is apparent now, however, that this symmetrization is necessitated *not* by considera-

tions of parity, but by considerations of exchange of identical nuclei.

Finally one can construct total wave functions as products of the rotation-nuclear spin functions $|JKM\epsilon\rangle$, and vibration functions $|\pm\rangle$. From Eqs. (A10) and (A15) these behave under permutation operations as

$$(12) |JKM\epsilon\rangle |\pm\rangle = \pm \epsilon(-)^J |JKM\epsilon\rangle |\pm\rangle. \quad (A19)$$

But this total wave function must be antisymmetric, so

$$\pm \epsilon(-)^J = -1 \quad \text{or} \quad \epsilon = \mp(-)^J, \quad (A20)$$

where the upper sign refers to $|\pm\rangle$, the lower level of each (J, K) doublet, and the lower sign refers to the upper level, $|\pm\rangle$. The parity of the levels is given from Eqs. (A3) and (A4) as

$$\begin{aligned} E^*(|JKM\rangle \mp (-)^J |J-KM\rangle) |\pm\rangle \\ = \pm (-)^K (|JKM\rangle \mp (-)^J |J-KM\rangle) |\pm\rangle, \end{aligned} \quad (A21)$$

which is found to agree with the parities as given, for example, in Refs. 16 and 17.

The final wave functions given by Eq. (A20) agree with the results quoted by Davis and Boggs.⁴ This is due, however, to a cancellation of two errors. A consistent derivation from Ref. 16 would require the correction in Eq. (A7), giving $\epsilon = \mp(-)^{J+K}$. But then one of the hydrogens is in the yz plane, and the form for the intermolecular potential used in Ref. 4 is improper. To maintain the form used for the potential, one of the hydrogens must be placed in the xz plane, in which case the effect of permuting hydrogen atoms, Eq. (A9), fortuitously agrees with the equation given incorrectly in Ref. 16, producing the apparent accord.

To complete this discussion, consider the effect of vibrational motion on matrix elements of the intermolecular potential. The potential $V(R, \Theta, \Phi, h)$ depends on the distance R , and orientation Θ and Φ , of the atomic projectile with respect to the molecule fixed axes, and also on the vibrational coordinate h . It is obvious from geometric considerations that⁸

$$V(R, \Theta, \Phi, -h) = V(R, \pi - \Theta, \Phi, h). \quad (A22)$$

If, following the discussion around Eq. (A2), the vibrational wave functions are assumed to be essentially delta functions, $\delta(h - h_e) \pm \delta(h + h_e)$, then, as noted by Davis and Boggs,⁴ matrix elements over vibrational motion are

$$\langle \pm | V | \pm \rangle = \{ V(R, \Theta, \Phi, h_e) + V(R, \Theta, \Phi, -h_e) \} / 2, \quad (A23a)$$

$$\langle \mp | V | \pm \rangle = \{ V(R, \Theta, \Phi, h_e) - V(R, \Theta, \Phi, -h_e) \} / 2. \quad (A23b)$$

If the potential at $h = h_e$ is expanded in spherical harmonics in the usual way

$$V(R, \Theta, \Phi, h_e) = \sum_{\lambda\mu} v_{\lambda\mu}(R) \{ Y_{\lambda\mu}(\Theta\Phi) + (-)^\mu Y_{\lambda-\mu}(\Theta\Phi) \}, \quad (A24)$$

one can use the fact that

$$Y_{\lambda\mu}(\pi - \Theta, \Phi) = (-)^{\lambda+\mu} Y_{\lambda\mu}(\Theta, \Phi) \quad (A25)$$

to get

$$\langle \pm | V | \pm \rangle = \sum_{\lambda\mu} v_{\lambda\mu} \{ Y_{\lambda\mu} + (-)^\mu Y_{\lambda-\mu} \} \{ 1 + (-)^{\lambda+\mu} \} / 2, \quad (A26a)$$

$$\langle \mp | V | \pm \rangle = \sum_{\lambda\mu} v_{\lambda\mu} \{ Y_{\lambda\mu} + (-)^{\mu} Y_{\lambda-\mu} \} \{ 1 - (-)^{\lambda+\mu} \} / 2. \quad (\text{A26b})$$

That is, for lower-to-lower and upper-to-upper matrix elements, only even $\lambda + \mu$ terms contribute; and for lower-to-upper matrix elements, only odd $\lambda + \mu$ contribute. Combining Eqs. (A20) and (A26) it is seen that nonzero matrix elements require

$$\epsilon \epsilon' (-)^{J+J'+\lambda+\mu} = +1. \quad (\text{A27})$$

But this selection rule is not new, having been obtained already in Ref. 8 [Eq. (46)] as a result of the "symmetrization" of the wave function, i.e., forming symmetric and antisymmetric combinations of $\pm K$.

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